

Notes of Learning Theory

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Outline

Intro - Learning Theory

Goal: Function Learning

Measurements: Excess Risk or Mean of Excess Risk

Method: Bounding the Generalization Gap

Tools

Dudley Inequalities

Symmetrization and Rademacher complexity

Uniform law of large numbers

VC law of large numbers

Wasserstein law of large numbers

Binary Classification

0-1 loss

cross-entropy loss

Linear Regression

Discussion

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Intro - Learning Theory

Recall classic statistics: parameter estimation (e.g. estimate the mean of a certain distribution)

- ▶ data: $\{x_i\}_{i=1}^n$
- ▶ hypothesis: distribution model of X
- ▶ method: statistics (e.g. $\hat{\theta} = \bar{X}$)
- ▶ analyze/evaluate: properties of the statistics ($\mathbb{E}(\hat{\theta} - \theta)$, ...) with $n/?$

Now in supervised learning: function estimation, i.e. parameter estimation under fixed functional form (e.g.)

- ▶ data: $\{(x_i, y_i)\}_{i=1}^n$
- ▶ hypothesis: distribution model of (X, Y) , $f : \mathcal{X} \rightarrow \mathcal{Y}$, $f \in \mathcal{F}$
- ▶ method: empirical risk minimization or ...

$$\mathcal{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i), \quad \hat{f} = \arg \min_{f \in \mathcal{F}} \mathcal{R}_n(f)$$

- ▶ analyze/evaluate: excess risk with n, m, T

$$\mathcal{R}(\hat{f}) - \mathcal{R}(f^*).$$

More on Learning theory, see xx

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Measurements of a function estimation

How well model f approximates pair r.v. $\{(x, y)\}$

Let's first consider the question: how to qualify that **how well a model $f : \mathcal{X} \rightarrow \mathcal{Y}$ approximates the data (distribution of the pair r.v. z)?**

Given (1) the (empirical or population) distribution of a two-dimension pair r.v. $z = (x, y)$, (2) a loss function $\ell : \mathcal{F} \times \mathcal{Z} \rightarrow \mathbb{R}_{\geq 0}$, $\ell(f, z) := \ell(f(x), y)$

► empirical risk

$$\mathcal{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f, z_i)$$

► population risk

$$\mathcal{R}(f) = \mathbb{E}_{z \sim \mu}[\ell(f, z)]$$

Examples of loss function

- Mean Squared Loss: $\ell(f, (x, y)) = (f(x) - y)^2$
- 0-1 Loss: $\ell(f, (x, y)) = \mathbb{1}\{f(x) \neq y\}$
- Cross Entropy: $\ell(f, (x, y)) = -y \log(f(x)) - (1 - y) \log(1 - f(x))$

Note that the population risk equals $\mathbb{E}[\mathcal{R}_n(f)]$ if we consider z to be an $n * 2$ -dimension r.v.[?]

Measurements of a function estimation

How well f_1 approximates f_2

Next: how to qualify **how well f_1 approximates f_2** ?

Moreover given (3) another model $f_2 : \mathcal{X} \rightarrow \mathcal{Y}$

- ▶ difference of population risks

$$\mathcal{R}(f_1) - \mathcal{R}(f_2) = \mathbb{E}_{z \sim \mu}[\ell(f_1, z) - \ell(f_2, z)]$$

which can be decomposed by

$$\begin{aligned}\mathcal{R}(f_1) - \mathcal{R}(f_2) &= [\mathcal{R}(f_1) - \mathcal{R}_n(f_1)] + [\mathcal{R}_n(f_1) - \mathcal{R}_n(f_2)] + [\mathcal{R}_n(f_2) - \mathcal{R}(f_2)] \\ &= \text{gen-gap}(f_1) + [\mathcal{R}_n(f_1) - \mathcal{R}_n(f_2)] - \text{gen-gap}(f_2)\end{aligned}$$

where the generalization gap is defined as

$$\text{gen-gap}(f) := \mathcal{R}(f) - \mathcal{R}_n(f) = \mathbb{E}_z[\ell(f, z)] - \frac{1}{n} \sum_{i=1}^n \ell(f, z_i)$$

Excess Risk

How well f approximates f^*

Suppose now we only consider function that is in a function class $\mathcal{F}$ (why?) and consider two special function, let f_1 and f_2 be

- ▶ $\hat{f}$ by empirical risk minimization $\hat{f} \in \arg \min_{f \in \mathcal{F}} \mathcal{R}_n(f)$
- ▶ f^* target model

The thing of interest in statistical learning is to analyze the behavior of the difference of population risks, which is called the **Excess Risk**.

$$\text{Excess Risk} = \mathcal{R}(\hat{f}) - \mathcal{R}(f^*)$$

Note that the Excess Risk is a random variable, since $\hat{f}$ is dependent on the dataset. So another form of goal is to analyze $\mathbb{E}(\mathcal{R}(\hat{f})) - \mathcal{R}(f^*)$, which is the mean of excess risk.

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Uniform bound of gen gap

The Excess Risk can be decomposed by

$$\begin{aligned}\mathcal{R}(\hat{f}) - \mathcal{R}(f^*) &= \text{gen-gap}(\hat{f}) + [\mathcal{R}_n(\hat{f}) - \mathcal{R}_n(f^*)] - \text{gen-gap}(f^*) \\ &\leq \text{gen-gap}(\hat{f}) - \text{gen-gap}(f^*) \\ &\leq 2 \sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)|\end{aligned}$$

The question boils down to

→ the behavior of the uniform bound of gen-gap

$$\sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)| = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \ell(f, z_i) - \mathbb{E}_{z \sim \mu} [\ell(f, z)] \right|$$

Expected result:

A bound on $\sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)|$ in terms of $n, \mathcal{F}$, conditional on μ, ℓ .

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Concentration and Suprema

Expected result:

A bound on $\sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)|$ in terms of $n, \mathcal{F}$, conditional on μ, ℓ .

Road Map:

1. Concentration: Consider $\sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)| = g(Z_1, \dots, Z_n)$, then by concentration inequalities, we can analyze its fluctuations around its mean. [if the function is fluent enough]
2. Suprema: Define random process $\mathbb{X}_f := \frac{1}{n} \sum_{i=1}^n \ell(f, Z_i) - \mathbb{E}[\ell(f, Z)]$, then by theory of suprema, we can analyze its expectation through the properties of the function class.

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Dudley for subgaussian-increments process

Let $(X_t)_{t \in T}$ be a mean-zero random process on a metric space (T, d) with subgaussian increments. Then

[Dudley, discrete inequality]

$$\mathbb{E} \sup_{t \in T} X_t \leq CK \sum_{k \in \mathbb{Z}} 2^{-k} \sqrt{\log \mathcal{N}(T, d, 2^{-k})}.$$

[Dudley, integral inequality]

$$\mathbb{E} \sup_{t \in T} X_t \leq CK \int_0^\infty \sqrt{\log \mathcal{N}(T, d, \varepsilon)} d\varepsilon.$$

Function space with subgaussian increments?



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Symmetrization

[Symmetrization for empirical processes] Let $\mathcal{F}$ be a class of Boolean functions on some domain Ω , and let $X, X_1, X_2, \dots, X_n$ be independent random points in Ω with common distribution. Then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| \leq 2 \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i f(X_i) \right|$$

where $\varepsilon_1, \varepsilon_2, \dots$ are independent Rademacher random variables, which are also independent of $X_1, X_2, \dots$.

A typical application of symmetrization

- ▶ 1. replace random variables X_i with symmetric ones $\varepsilon_i X_i$
- ▶ 2. condition on X_i so that all randomness comes from Rademacher r.v.'s ε_i

Rademacher complexity

Rademacher variable ε ,

$$\mathbb{P}(\varepsilon = 1) = \mathbb{P}(\varepsilon = -1) = \frac{1}{2}$$

The Rademacher complexity of a set $A \subseteq \mathbb{R}^n$

$$\text{Rad}(A) = \mathbb{E}_{\varepsilon} \left(\sup_{\alpha \in A} |\langle \alpha, \varepsilon \rangle| \right)$$

Empirical version

Characteristics of Rad?

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VC dimension and Covering Number

VC law of large numbers

Boolean function space

Bounding the suprema of empirical process indexed by Boolean $\mathcal{F}$ space

[VC law of large numbers] Consider the class of functions

$$\mathcal{F} := \{f : \Omega \rightarrow \{0, 1\}\},$$

with finite VC dimension. Let $X, X_1, X_2, \dots, X_n$ be i.i.d. random variables from distribution on Ω . Then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| \leq C \sqrt{\frac{\text{vc}(\mathcal{F})}{n}}.$$

Proof Roadmap

1. symmetrization and conditional on X_i , $\rightarrow \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i f(X_i) \right|$
2. Dudley $\rightarrow$ function of covering number;
3. take expectation w.r.t. X_i ;
4. covering number $\leftrightarrow$ VC dimension

Wasserstein law of large numbers

a.k.a Lipschitz law of large numbers

Bounding the suprema of empirical process indexed by Lipschitz $\mathcal{F}$ space

[Lipschitz law of large numbers] Consider the class of functions

$$\mathcal{F} := \{f : [0, 1] \rightarrow \mathbb{R}, \|f\|_{\text{Lip}} \leq L\},$$

where L is any number. Let $X, X_1, X_2, \dots, X_n$ be i.i.d. random variables taking values in $[0, 1]$. Then

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - \mathbb{E} f(X) \right| \leq \frac{CL}{\sqrt{n}}.$$

Proof Roadmap

1. Checking subgaussian w.r.t. L^∞
2. Dudley $\rightarrow$ function of covering number $\frac{1}{\sqrt{n}} \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon)} d\varepsilon$
3. Integral with $\mathcal{N}(\mathcal{F}, \|\cdot\|_{L^\infty}, \varepsilon) \leq e^{C/\varepsilon}$ since $\mathcal{F}$ is bounded

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Binary Classification

- ▶ data: $\{(x_i, y_i)\}_{i=1}^n, \sim \mu?$ (not necessarily)
- ▶ population: μ over $\mathcal{Z} = \mathcal{X} \times \{0, 1\}$
- ▶ function class: $\mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow \{0, 1\}\}$
- ▶ loss function: for $z = (x, y)$, $\ell(f, z) = \mathbb{1}\{f(x) \neq y\} \in [0, 1]$

Binary Classification

Suprema

By [Symmetrization of Empirical Processes]

$$\begin{aligned}\mathbb{E} \left[\sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)| \right] &\leq \frac{2}{n} \mathbb{E}_{Z, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \varepsilon_i \ell(f, Z_i) \right| \right] \\ &= \frac{2}{n} \mathbb{E}_{Z, \varepsilon} \left(\sup_{\alpha \in S(Z)} |\langle \alpha, \varepsilon \rangle| \right) \\ &\leq \frac{2}{n} \mathbb{E}_Z [\text{Rad}(S(Z))]\end{aligned}$$

where

$$S(Z) = \left\{ (\ell(f, Z_1), \dots, \ell(f, Z_n)) : f \in \mathcal{F} \right\} \subseteq [0, 1]^n$$

Binary Classification

Concentration

Since 0-1 loss is bounded, we have

$$\sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)| = g(Z_1, \dots, Z_n), \quad \|D_i g\|_\infty \leq \frac{2}{n},$$

Then by McDiarmid, $\forall t > 0$,

$$\mathbb{P}\{g(Z_1, \dots, Z_n) - \mathbb{E}[g(Z_1, \dots, Z_n)] \geq t\} \leq \exp(-nt^2/2).$$

Equivalently, $\forall \delta \in (0, 1)$,

$$\mathbb{P}(\sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)| \leq \mathbb{E} \left[\sup_{f \in \mathcal{F}} |\mathcal{R}_n(f) - \mathcal{R}(f)| \right] + \sqrt{\frac{2 \log(1/\delta)}{n}}) \geq 1 - \delta$$

Binary Classification

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OLS with fixed design

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- ▶ Parameterizing[?]
- ▶ fully determined labels $T(X)$ v.s. random and correlated (X, Y) : In this note we only talk about the case when data is of the form $\{x_i, y_i\}_{i=1}^n$, in (Vershynin 2026) there is the view of $\{x, T(X)\}_{i=1}^n$ (Exercise 8.30)

References

- Handel, Ramon van (2016). “Probability in High Dimension”. URL: <https://web.math.princeton.edu/~rvan/APC550.pdf>.
- Vershynin, Roman (2026). *High-Dimensional Probability: An Introduction with Applications in Data Science*. 2nd ed. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press.
- Wu, Lei (2026). *Topics in Deep Learning Theory*. Lecture Notes, Peking University. Accessed: 2026-06-29. URL: <https://leiwu0.github.io/courses/dl-theory/index.html>.